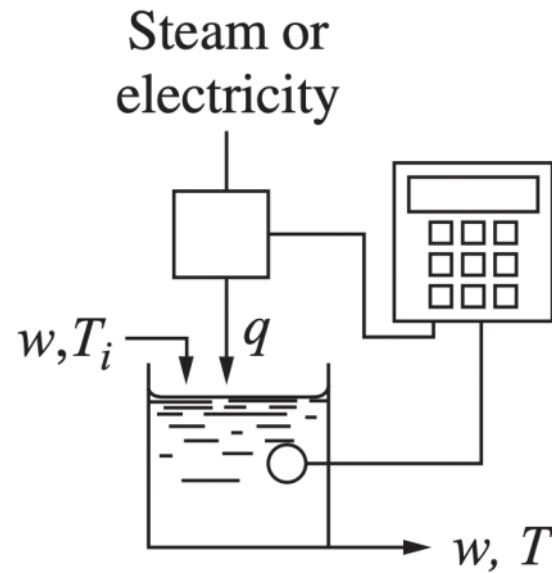


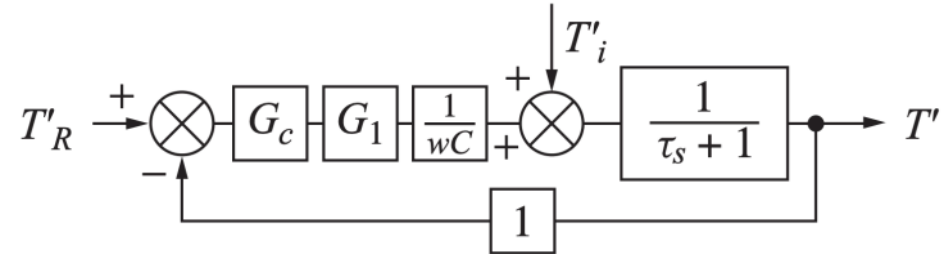
Transient Response of Simple Control Systems

→ Response of a simple control system to changes in set point and load



Heated Stirred tank

→ Heat can be provided by
 - Steam
 - Electricity

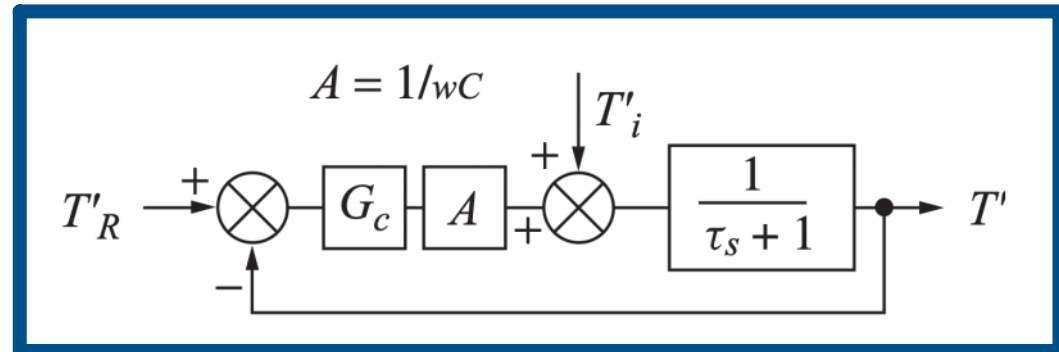


→ $A = \frac{1}{wC}$; $G_1 = K_c = \text{const}$

→ $K_c = 1 \Rightarrow \text{if } K_c \neq 1 ; G_c = G_{c0} K_v$

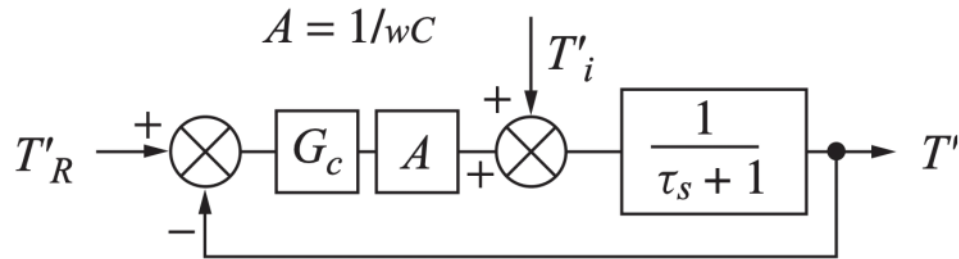
→ $H = 1 \Rightarrow \text{unity feedback system}$

Simplified
Block
Diagram



Proportional Control for Set Point change

→ $U=0 \Rightarrow$ Servo Problem



Overall Transfer function

$$\frac{T'}{T'_R} = \frac{G_c A / (\tau s + 1)}{1 + (G_c A / (\tau s + 1))}$$

→ For Proportional control
 $G_c = K_c$

$$\frac{T'}{T'_R} = \frac{\frac{K_c A}{\tau s + 1}}{1 + \frac{K_c A}{\tau s + 1}} = \frac{K_c A}{(\tau s + 1) \left(\frac{\tau s + 1 + K_c A}{\tau s + 1} \right)}$$

$$\frac{T'}{T'_R} = \frac{K_c A}{\tau s + (1 + K_c A)}$$

Rearrange to first order transfer func.

$$\frac{T'}{T'_R} = \frac{K_c A / (1 + K_c A)}{\left(\tau / (1 + K_c A) \right) s + 1}$$

$$\boxed{\frac{T'}{T'_R} = \frac{A_1}{\tau_1 s + 1}}$$

$$\tau_1 = \frac{\tau}{1 + K_c A}$$

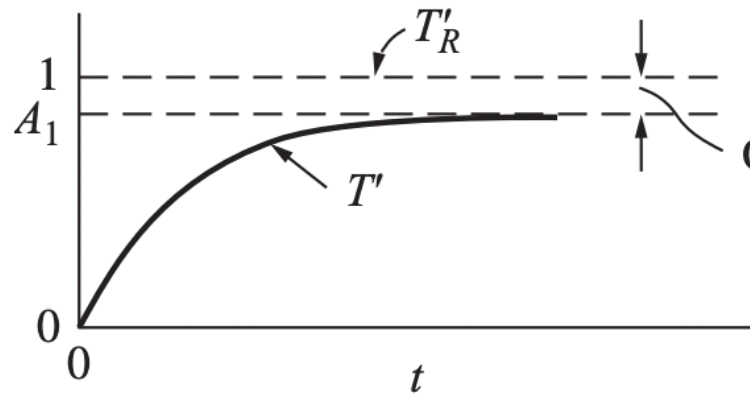
$$A_1 = \frac{K_c A}{1 + K_c A} = \frac{1}{1 + 1/K_c A}$$

$$\frac{I'}{I_R} = \frac{A_1}{\tau_1 s + 1}$$

$$\tau_1 = \frac{\tau}{1 + K_c A} \Rightarrow \text{Time const of closed-loop control } \tau_1 < \tau$$

$\rightarrow$ Feedback Control speeds up response

$$A_1 = \frac{K_c A}{1 + K_c A} = \frac{1}{1 + 1/K_c A}$$



$$\text{Offset} = T'_R(\infty) - T'(\infty)$$

$$= 1 - \frac{K_c A}{1 + K_c A} = \frac{1 + K_c A - K_c A}{1 + K_c A} = \frac{1}{1 + K_c A}$$

For proportional control
as $K_c \uparrow$ Offset $\downarrow$

FIGURE 12-2

Unit-step response for set point change
(P control).

$$T'_R(\infty) = 1$$

Desired

$$T' \sim A_1 = K_c A / (1 + K_c A)$$

Acquired

Example 12.1

Stirred tank heaters

Determine the response of a closed loop proportional control system for $K_c = 5, 10, 20$ and 100

Set point change = 5°C

Determine offset for each value of K_c

Deviation Variables:-

$$T_i' = T_i - T_{is} = T_i - 60$$

$$T_R' = T_R - T_{Rs} = T_R - 80$$

$$q' = q - q_s = q - 280 \text{ kW}$$

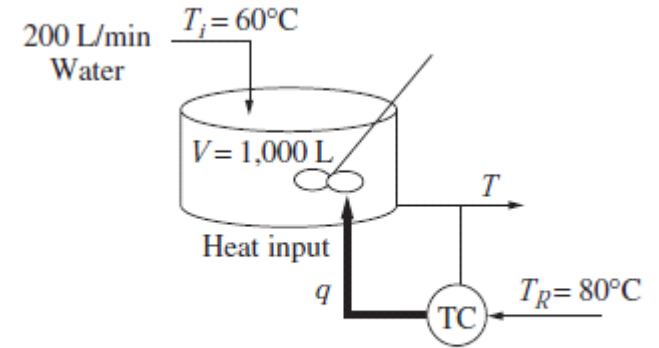


FIGURE 12-3
Stirred-tank heater control system.

$$\tau = \frac{\text{Volume}}{\text{Volumetric flow rate}} = \frac{1000 \text{ L}}{200 \text{ L/min}} = 5 \text{ min}$$

$$A = \frac{1}{WC} = \frac{1^\circ\text{C}}{14 \text{ kW}} = 0.0174^\circ\text{C/kW}$$

Both calculated before

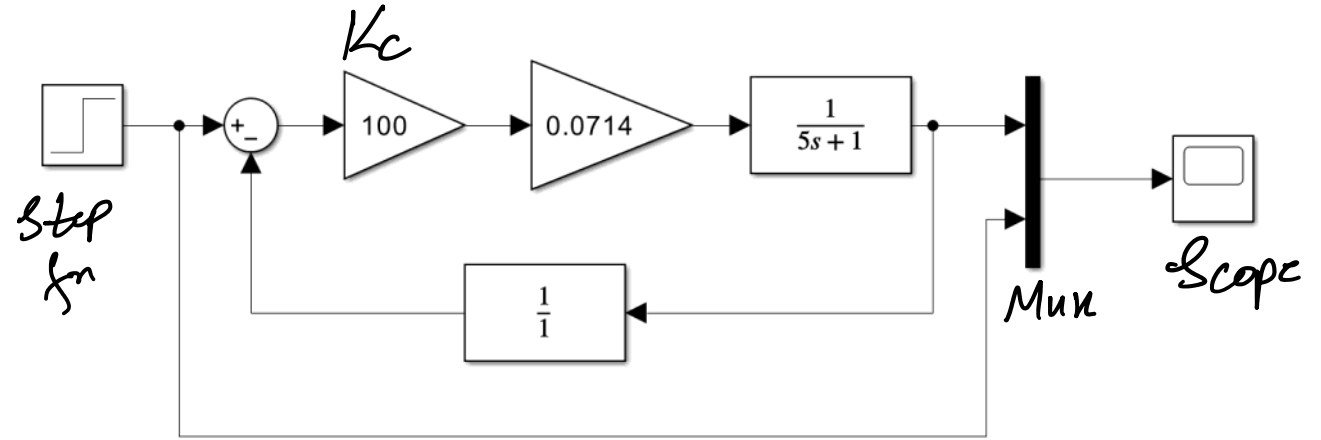
$$Offset = \frac{1}{1+K_c A}$$

for

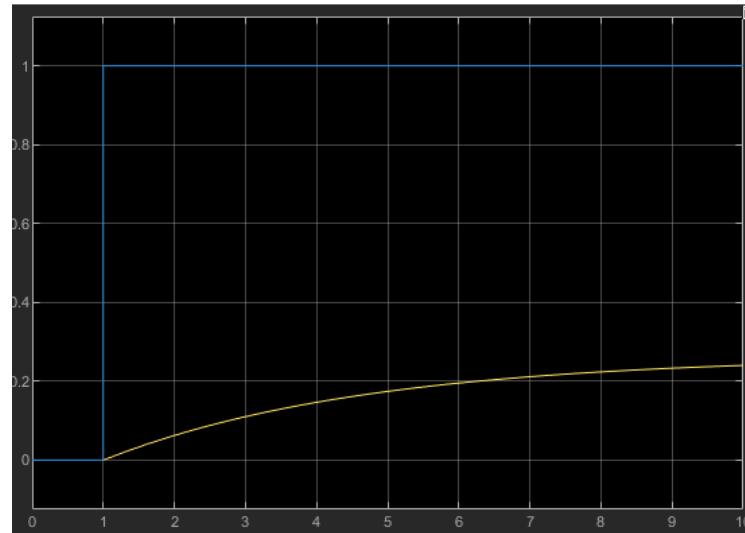
$$K_c = 5 \quad \frac{1}{1+(5)(0.0714)} = 3.69$$

$$\vdots$$

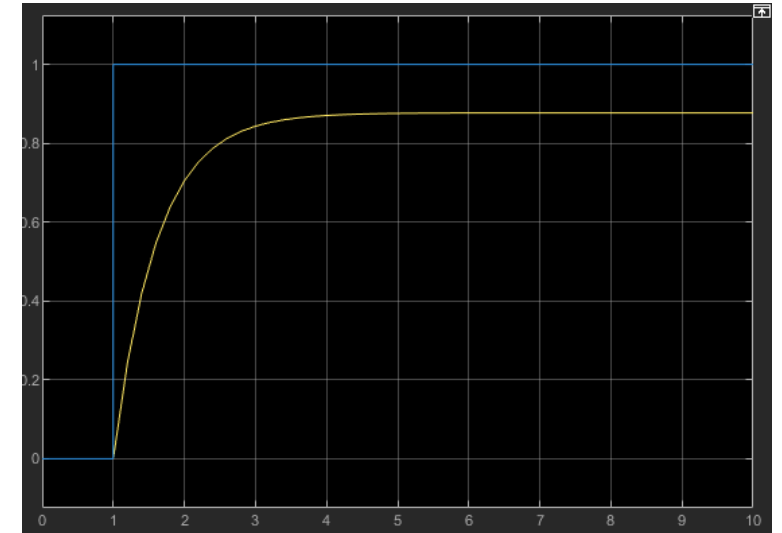
$$K_c = 100 \quad \frac{1}{1+(100)(0.0714)} = 0.614$$



$K_c = 5$



$K_c = 100$



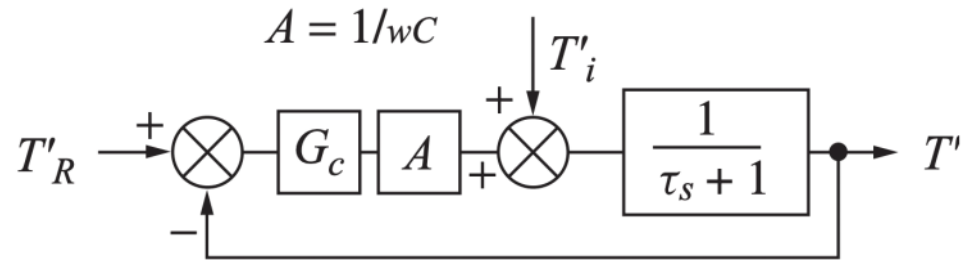
Proportional Control for Load change

⇒ Regulator Problem ⇒ Disturbance Rejection

$$T'_R = 0$$

⇒ Inlet stream temp changes

We desire $T'(\infty) = 0$



Overall Transfer function

$$\frac{T'}{T'_i} = \frac{1/\tau_s + 1}{1 + \frac{K_c A}{\tau_s + 1}} = \frac{1}{(\tau_s + 1) + K_c A}$$

$$\frac{T'}{T'_i} = \frac{1}{\tau_s + 1 + K_c A} = \frac{\frac{1}{1 + K_c A}}{\frac{\tau_s}{1 + K_c A} + \frac{1 + K_c A}{1 + K_c A}} = \frac{A_2}{\tau_1 s + 1}$$

$$A_2 = \frac{1}{1 + K_c A}$$

$$\tau_1 = \frac{\tau}{1 + K_c A}$$

For step change in T_i

$$T_i = \frac{1}{s}$$

$$T' = \frac{1}{s} \frac{\frac{1}{1+K_c A}}{\left(\frac{2}{1+K_c A}\right)s + 1}$$

Ultimate s.s value $T'(\infty) = 0$

$$\lim_{s \rightarrow 0} s T'(s) = s \frac{1}{s} \frac{\frac{1}{1+K_c A}}{\left(\frac{2}{1+K_c A}\right)s + 1}$$

$$\lim_{s \rightarrow 0} s T(s) = \frac{1}{1+K_c A}$$

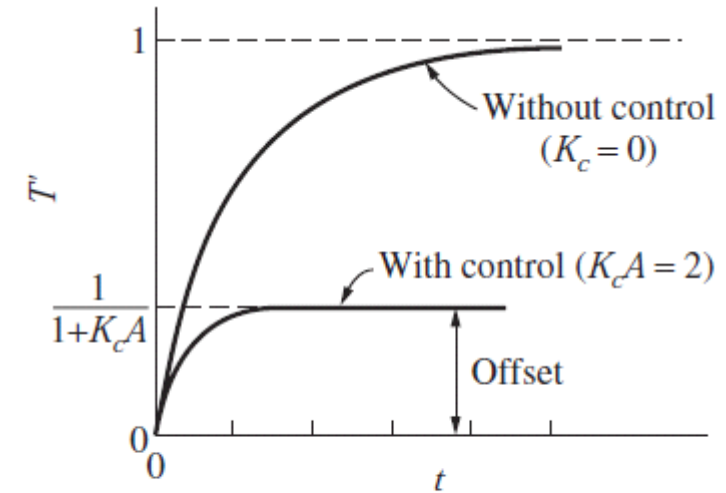


FIGURE 12-7
Unit-step response for load change (P control).

$$\begin{aligned} \text{Offset} &= T_R'(\infty) - T'(\infty) \\ &= 0 - \frac{1}{K_c A + 1} \end{aligned}$$

$$\text{Offset} = -\frac{1}{K_c A + 1}$$

Example 12.2
 Proportional control of a Stirred Tank Heater for Disturbance Rejection
 for system in Example 12.1
 disturbance of 5°C
 Offset for each value of K_c
 $K_c = 5, 10, 20, 100$

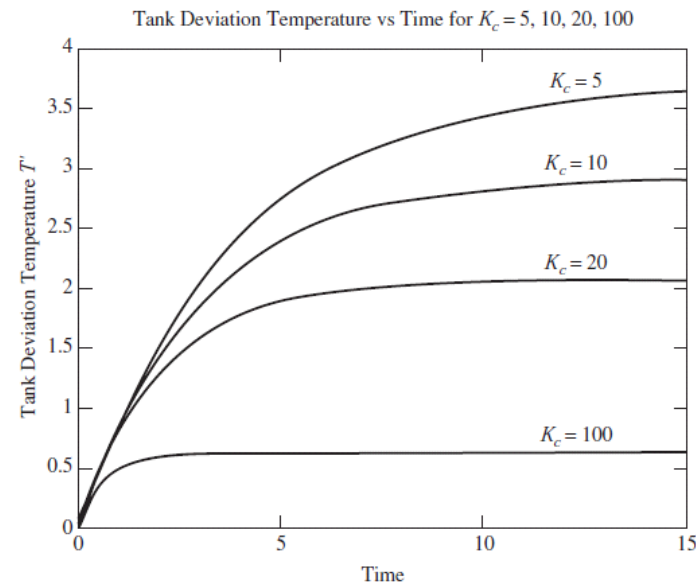


FIGURE 12-8
 Response of stirred-tank temperature for proportional control—disturbance rejection.

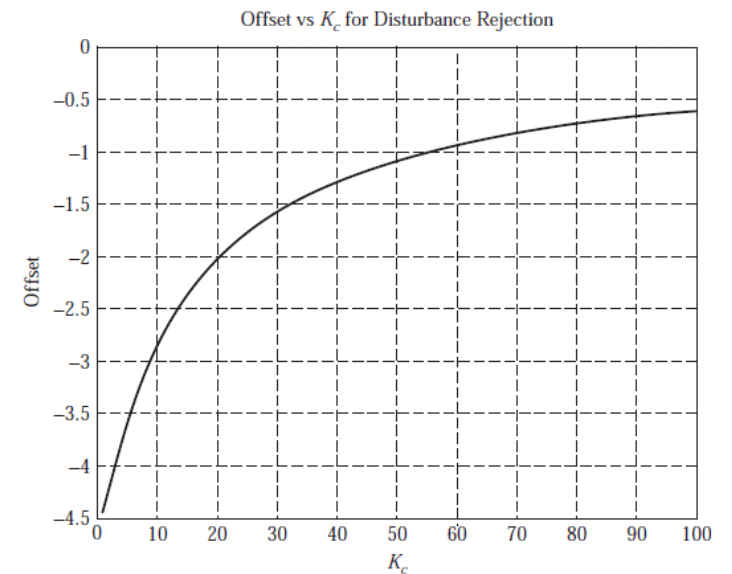
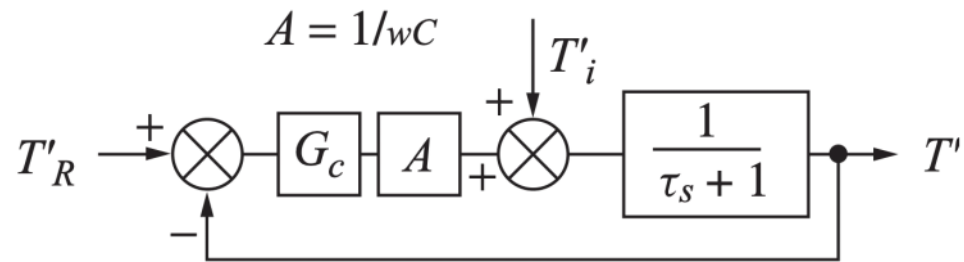


FIGURE 12-9
 Effect of controller gain on offset for stirred-tank proportional control—disturbance rejection.

Proportional Integral Control for Load Change

Goal is disturbance rejection



For PI control

G_c replaced with $K_c (1 + \frac{1}{\tau_I s})$

Overall Transfer function

$$\frac{T'}{T'_i} = \frac{1/(\tau_s + 1)}{1 + \frac{K_c (1 + \frac{1}{\tau_I s}) A}{\tau_s + 1}}$$

$$\frac{T'}{T'_i} = \frac{1}{(\tau_s + 1) (\tau_s + 1) + K_c (1 + \frac{1}{\tau_I s}) A}$$

$$= \frac{1}{(\tau_s + 1) + K_c \left(\frac{\tau_I s + 1}{\tau_I s} \right) A}$$

$$= \frac{\tau_I s}{(\tau_s + 1) (\tau_I s) + K_c (\tau_I s + 1) A}$$

$$= \frac{\tau_I s}{\tau \tau_I s^2 + \tau_I s + K_c A \tau_I s + K_c A}$$

$$= \frac{\tau_I s}{\tau \tau_I s^2 + s(\tau_I + K_c A \tau_I) + K_c A}$$

Converting to Standard Second Order function

$$\frac{T'}{T_i'} = \frac{(\tau_I s) / K_c A}{\frac{\tau \tau_I}{K_c A} s^2 + s \left(\frac{\tau_I}{K_c A} + \tau_I \right) + 1}$$

$$= \frac{A_1 s}{\tau_1^2 s^2 + 2 \zeta \tau_1 s + 1}$$

$$A_1 = \frac{\tau_I}{K_c A}$$

$$\tau_1 = \sqrt{\frac{\tau \tau_I}{K_c A}}$$

$$\zeta = \frac{1}{2} \sqrt{\frac{\tau_I}{\tau}} \frac{1 + K_c A}{\sqrt{K_c A}}$$

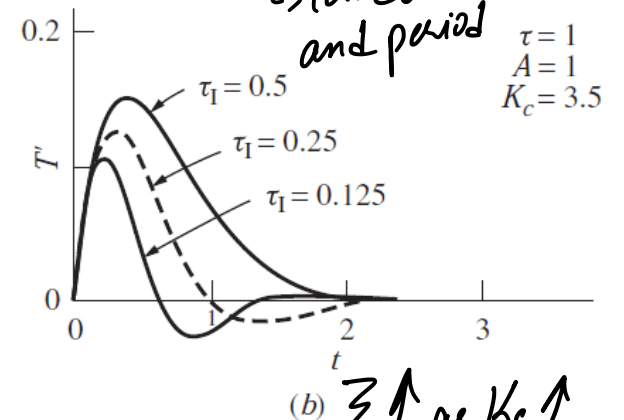
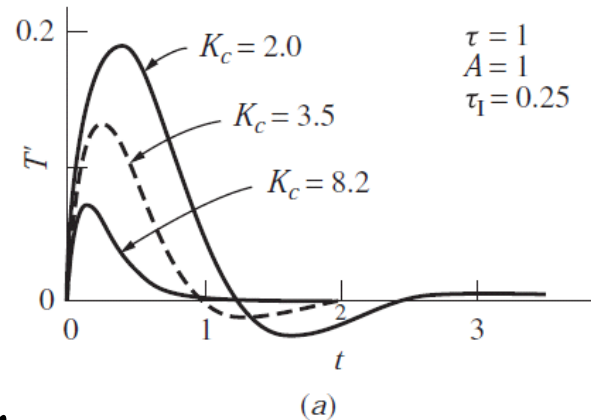
$K_c \uparrow$
→ less deviation

Impulse Response

$$y(t) = \frac{1}{2} \frac{1}{\sqrt{1-\zeta^2}} e^{-\frac{\zeta t}{\tau_1}} \sin \sqrt{1-\zeta^2} \frac{t}{\tau_1}$$

$$T' = A_1 \left[\frac{1}{\tau_1} \frac{1}{\sqrt{1-\zeta^2}} e^{-\frac{\zeta t}{\tau_1}} \sin \sqrt{1-\zeta^2} \frac{t}{\tau_1} \right]$$

as $\tau_I \downarrow$
→ less deviation and period



$\zeta \uparrow$ as $K_c \uparrow$
→ less oscillation

FIGURE 12-10

Unit-step response for load change—disturbance rejection (PI control).

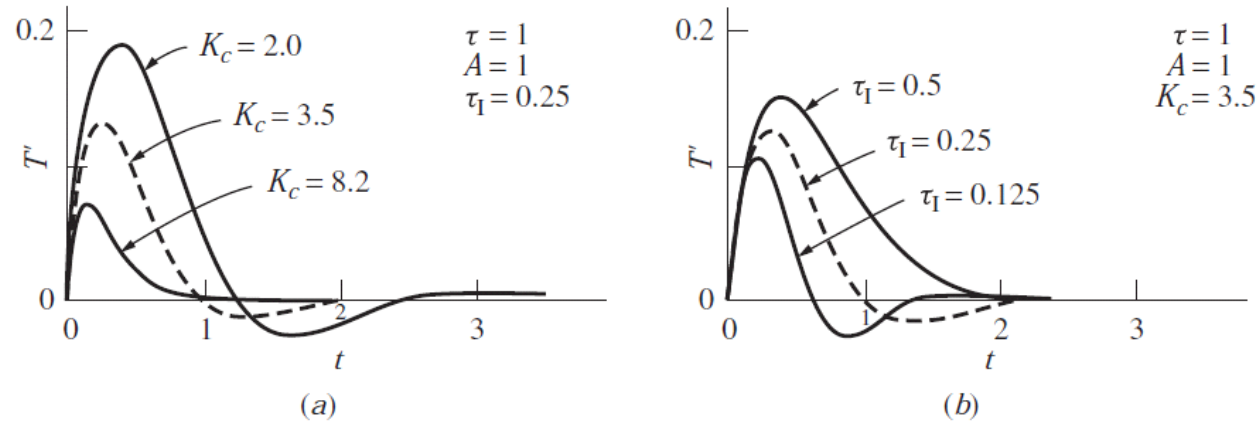


FIGURE 12-10
Unit-step response for load change—disturbance rejection (PI control).

Summary for Proportional-Integral Control—Response to Step Change in Load

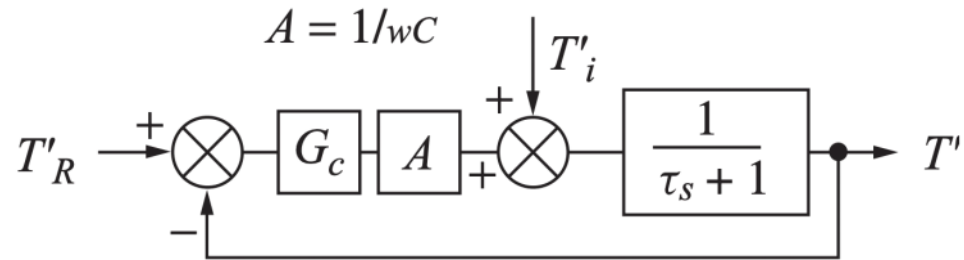
For fixed value of τ_I : As $K_c \uparrow$, $\zeta \uparrow$	Max. deviation $\downarrow$	Oscillations $\downarrow$
For fixed value of K_c : As $\tau_I \downarrow$, $\zeta \downarrow$	Max. deviation $\downarrow$	Oscillations $\uparrow$

For PI control $\Rightarrow \text{Offset} = T_R'(\infty) - T(\infty) = 0 - 0 = 0$
 $\Rightarrow \text{No Offset}$

Example 12.3. Proportional-integral control of a stirred tank heater for disturbance rejection.

- (a) Determine the response of a closed-loop proportional control system in Example 12.1 for $K_c = 5, 10, 20$, and 100 for a disturbance of 5°C (i.e., the feed temperature increases by 5°C from 60 to 65°C). Use a value of 2 min for τ_I .
- (b) Repeat part (a), using a constant value of 20 for K_c and values of $1, 2, 5$, and 10 min for τ_I .

Proportional Integral Control for Set Point Change



$$G_c = K_c \left(1 + \frac{1}{\tau_I s} \right)$$

$$\begin{aligned} \frac{T'}{T'_R} &= \frac{A K_c \left(1 + \frac{1}{\tau_I s} \right) \left(\frac{1}{\tau_s + 1} \right)}{1 + A K_c \left(1 + \frac{1}{\tau_I s} \right) \left(\frac{1}{\tau_s + 1} \right)} \\ &= \frac{A K_c \frac{\tau_I s + 1}{\tau_I s} \frac{1}{\tau_s + 1}}{1 + A K_c \frac{\tau_I s + 1}{(\tau_s + 1) \tau_I s}} \end{aligned}$$

$$\frac{T'}{T'_R} = \frac{A K_c (\tau_I s + 1)}{(\tau_s + 1) \tau_I s + A K_c (\tau_I s + 1)}$$

$$= \frac{\tau_I s + 1}{(\tau_s + 1) \tau_I s + \tau_I s + 1} \frac{A K_c}{A K_c}$$

$$= \frac{\tau_I s + 1}{\tau \tau_I s^2 + \tau_I s + \tau_I s + 1} \frac{A K_c}{A K_c}$$

$$\Rightarrow \tau_1 = \sqrt{\frac{\tau \tau_I}{K_c A}} ; \quad \zeta = \frac{1}{2} \sqrt{\frac{\tau_I}{\tau}} \frac{1 + K_c A}{\sqrt{K_c A}}$$

$$\Rightarrow \frac{\tau_I s}{A K_c} + \tau_I s$$

$$= \frac{\tau_I s + K_c A \tau_I s}{K_c A}$$

$$= \frac{\tau_I s (1 + K_c A)}{K_c A} \times \frac{\tau_1}{\tau_1}$$

$$= \frac{\tau_I (1 + K_c A)}{K_c A} \sqrt{\frac{K_c A}{\tau \tau_I}} \tau_1 s$$

$$= \sqrt{\frac{\tau_I}{\tau}} \frac{1+K_c A}{\sqrt{K_c A}} \tau_I s \times \frac{2}{2}$$

$$= \left(\frac{1}{2} \sqrt{\frac{\tau_I}{\tau}} \frac{1+K_c A}{\sqrt{K_c A}} \right) \tau_I s \quad 2$$

$$= \tau_I s$$

⇒ Plugging back

$$\frac{T'}{T_R} = \frac{\tau_I s + 1}{\tau_I^2 s^2 + 2 \tau_I s + 1}$$

For Step change in set point

$$T' = \frac{1}{s} \frac{\tau_I s + 1}{\tau_I^2 s^2 + 2 \tau_I s + 1}$$

$$= \frac{\tau_I}{\tau_I^2 s^2 + 2 \tau_I s + 1} + \frac{1}{s} \frac{1}{\tau_I^2 s^2 + 2 \tau_I s + 1}$$

Unit
step
response

Impulse
response

$$Y(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta t/\tau} \sin \left(\sqrt{1-\zeta^2} \frac{t}{\tau} + \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta} \right) \quad (7.18)$$

$$Y(t) = \frac{1}{\tau} \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta t/\tau} \sin \sqrt{1-\zeta^2} \frac{t}{\tau} \quad (7.32)$$

$$T' = \left(\frac{\tau_I}{\tau_I \sqrt{1-\zeta^2}} e^{-\frac{\zeta t}{\tau_I}} \sin \sqrt{1-\zeta^2} \frac{t}{\tau_I} \right) + 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\frac{\zeta t}{\tau_I}} \sin \left(\sqrt{1-\zeta^2} \frac{t}{\tau_I} + \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta} \right)$$

Standard Second Order Transfer Function

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad ; \quad \omega_n = \frac{1}{T}$$

$$= \frac{1}{T^2 s^2 + 2\zeta T s + 1}$$

Response for under damped $\zeta < 1$ Step change

$$y(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} \cdot t + \phi)$$

$$\phi = \tan^{-1} \left(\frac{\sqrt{1-\zeta^2}}{\zeta} \right)$$

Impulse

$$y(t) = \omega_n \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} t)$$

$$\text{Offset} = T_R'(\infty) - T'(\infty)$$

$$= 1 - 1 = 0$$

Integral Element Eliminates Offset

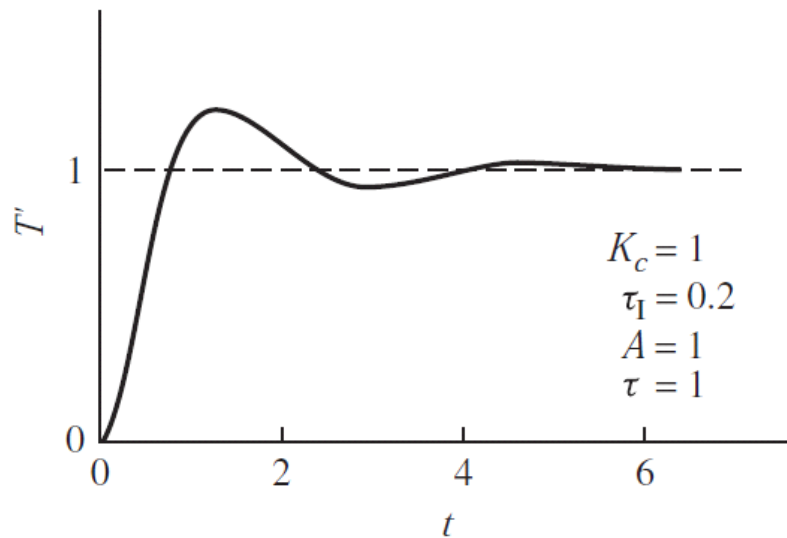


FIGURE 12-14

Unit-step response for set point change (PI control).

Example 12.4. Proportional-integral control of a stirred-tank heater for set point tracking

- Determine the response of a closed-loop proportional control system in Example 12.1 for $K_c = 5, 10, 20$, and 100 for a set point change of 5°C . Use a value of 2 min for τ_I .
- Repeat part (a) using a constant value of 20 for K_c and values of $1, 2, 5$, and 10 min for τ_I .

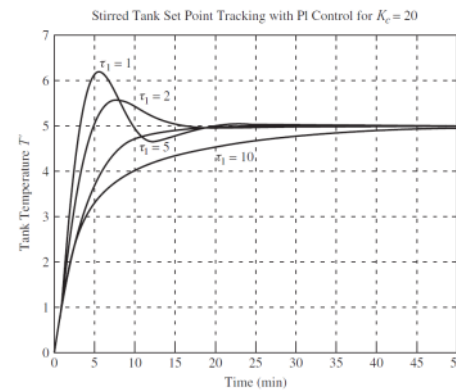
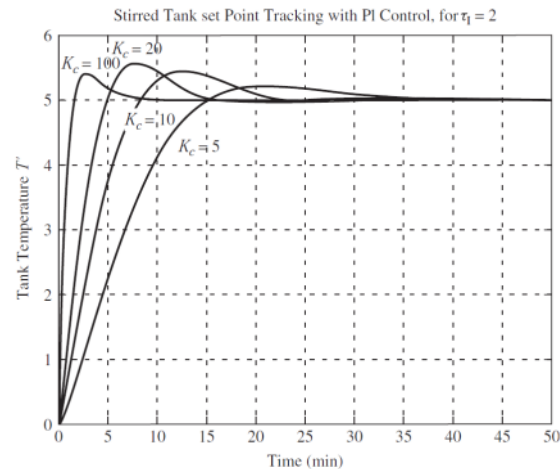
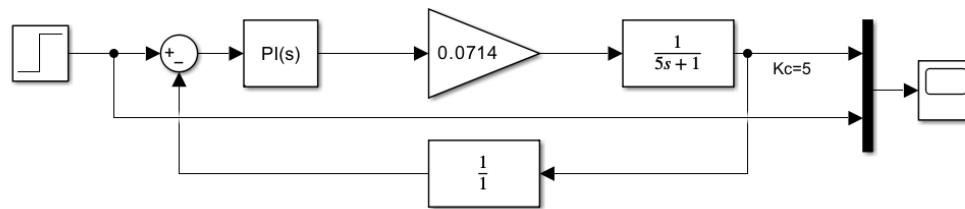
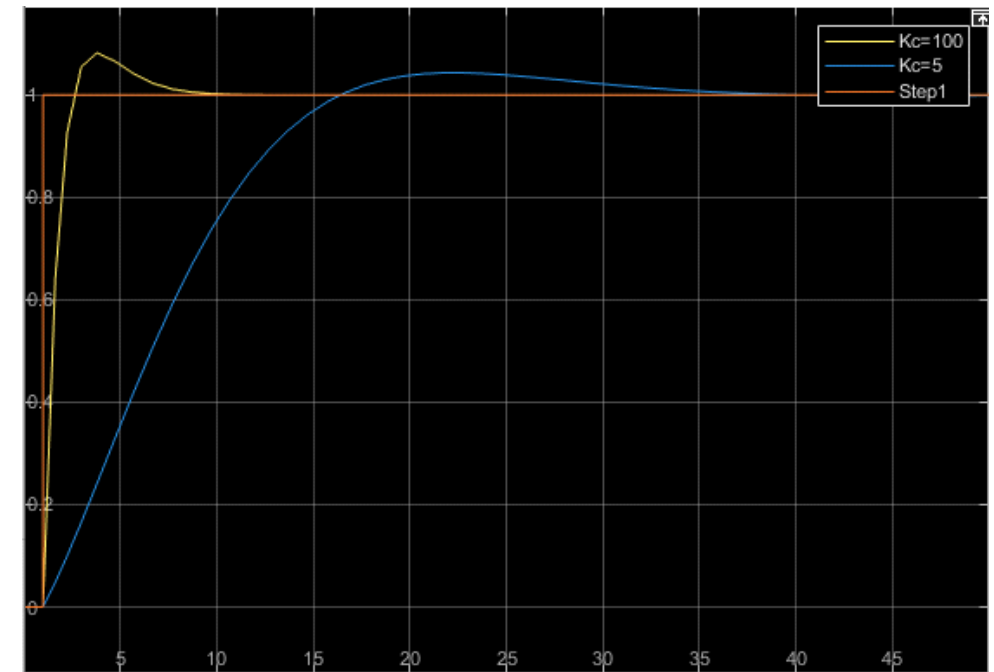


FIGURE 12-16 Stirred-tank set point tracking with PI control ($K_c = 20$).



Proportional Control of System with Measurement Lag

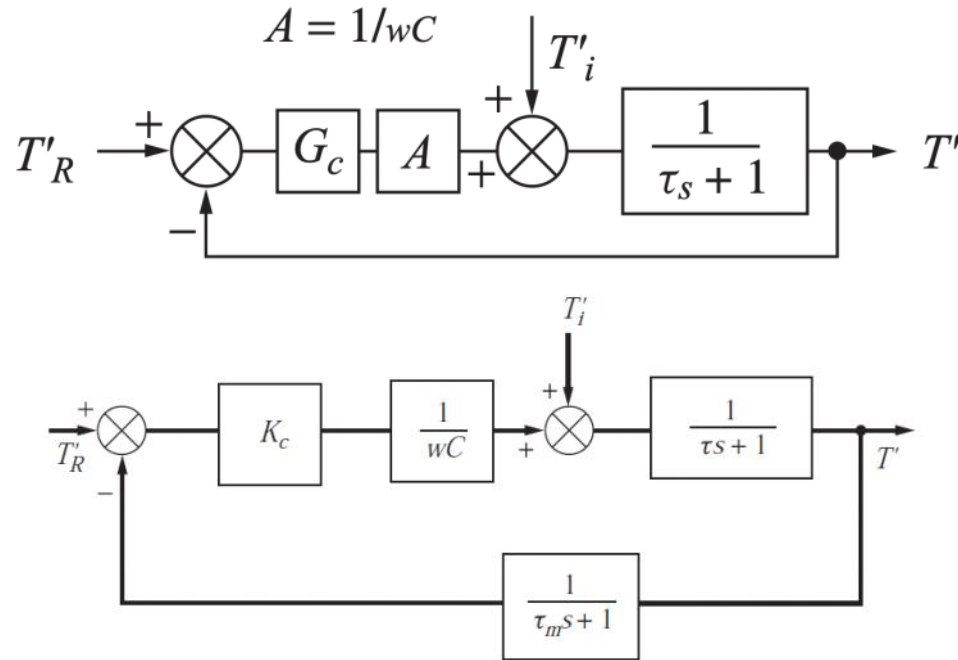


FIGURE 12-17
Control system with measurement lag.

Now considering measurement lag
as first order $\frac{1}{\tau_m s + 1}$

For set point change
With Overall transfer function as

$$\begin{aligned}
 \frac{T'}{T'_R} &= \frac{K_c A / \tau_s s + 1}{1 + \frac{K_c A}{(\tau_s s + 1)(\tau_m s + 1)}} \\
 &= \frac{\cancel{K_c A} / \tau_s s + 1}{(\tau_s s + 1)(\tau_m s + 1) + \cancel{K_c A}} \\
 &= \frac{K_c A (\tau_m s + 1)}{\tau_s \tau_m s^2 + \tau_s s + \tau_m s + 1 + K_c A} \\
 &= \frac{K_c A (\tau_m s + 1)}{\tau_s \tau_m s^2 + (\tau_s + \tau_m) s + (1 + K_c A)}
 \end{aligned}$$

$$\frac{T'}{T_R'} = \frac{\frac{K_c A}{(1+K_c A)} (\tau_m s + 1)}{\frac{\tau \tau_m}{1+K_c A} s^2 + \frac{\tau + \tau_m}{1+K_c A} s + 1} = \frac{A_1 (\tau_m s + 1)}{\tau_2^2 s^2 + 2 \zeta_2 \tau_2 s + 1}$$

$$A_1 = \frac{K_c A}{1+K_c A}$$

$$\tau_2 = \sqrt{\frac{\tau \tau_m}{1+K_c A}}$$

$$\begin{aligned} \Rightarrow & \frac{\tau + \tau_m}{1+K_c A} \times \frac{\tau_2}{\tau_2} \\ &= \frac{\tau + \tau_m}{1+K_c A} \frac{\sqrt{1+K_c A}}{\sqrt{\tau \tau_m}} \tau_2 \times \frac{\tau_2}{2} \\ &= 2 \tau_2 \left(\frac{\tau + \tau_m}{\sqrt{1+K_c A}} \frac{1}{2 \sqrt{\tau \tau_m}} \right) = 2 \zeta_2 \tau_2 \end{aligned}$$

Example 12.5 Effect of measurement lag on the proportional-integral control of a stirred-tank heater for set point tracking. Let's examine the effect of measurement lag on the PI control of the stirred-tank heater. We just concluded that as measurement lag increases, a proportional control only system becomes more oscillatory. What happens with PI control? The Simulink model is modified to include measurement lag (see Fig. 12–19). We'll fix K_c at 20 and $\tau_I = 2$, which gave us reasonably good results in Example 12.4.

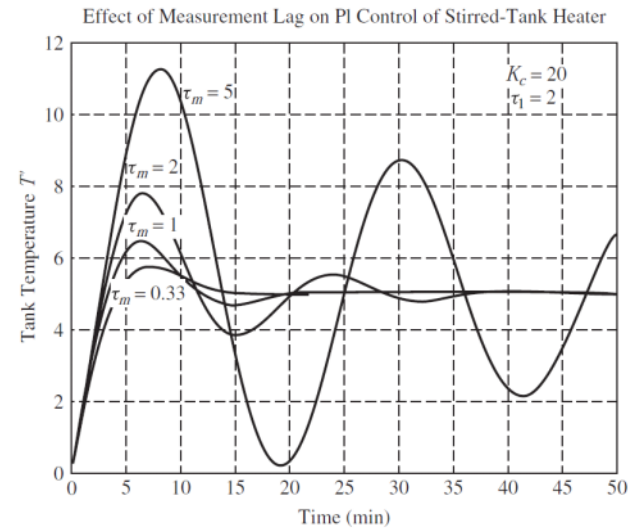
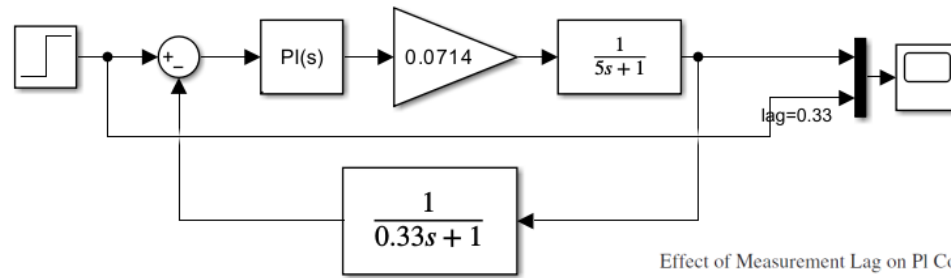
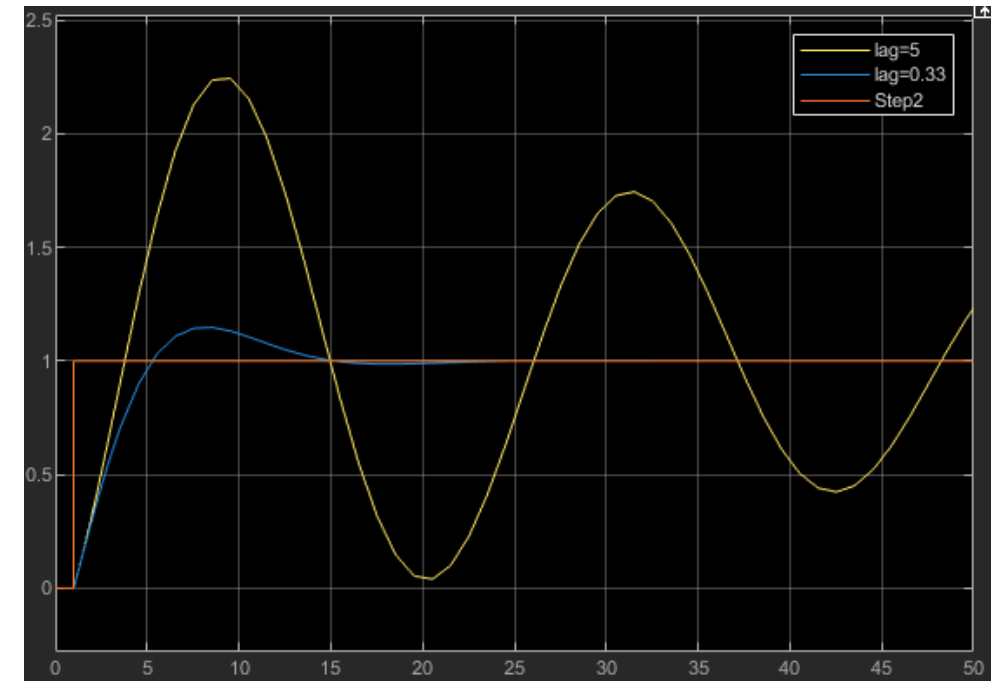
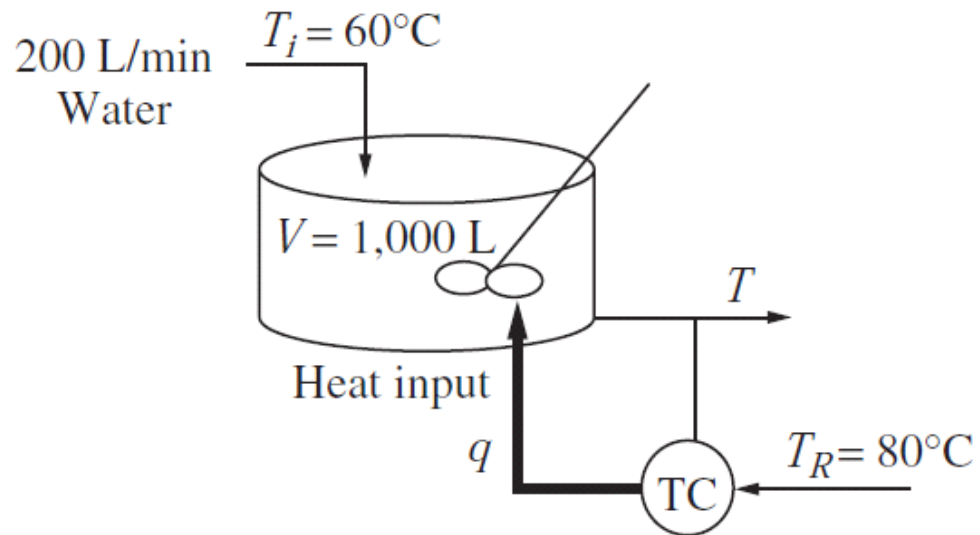


FIGURE 12–20
Effect of measurement lag on PI control of stirred-tank heater.



Homework 7

Name: _____



For the Stirred-tank heater shown here

- Determine the response of a closed-loop proportional control system for $K_c = 5, 10, 10$ and 100 for a set point change of 5°C . Plot the response for each case and show how the offset changes
- Determine the proportional response for $K_c = 5, 10, 10$ and 100 for a disturbance of 5°C . Plot the response and the offset.
- With PI control administered, determine the response for $K_c = 20$ for a disturbance of 5°C while using integral times of $1, 2, 5$, and 10
- With PI control administered, determine the response for $K_c = 20$ for a set point change of 5°C while using integral times of $1, 2, 5$, and 10
- What is the effect of changing measurement lag with constant value of $K_c = 20$ and integral time of 2

Tips:

Use Simulink for the solution.

For pasting pictures, try taking screenshots using "Windows+Shift+S" and pasting directly on here

For adding word or pdf documents, add them here and right click to select "Insert as Printout"

ANSWER HERE